

Name: _____

CWID: One digit per box.

Sheet 1 of 3

Practice Exam

ECON 201

Instructions

- 65 minutes. Three parts, weighted equally at 10 points each, each on its own sheet printed front and back: Part 1, ten multiple-choice questions; Part 2, short-answer problems; Part 3, one long-form question.
- Closed book and closed notes. A basic calculator is allowed. You will be given the help sheet separately.
- Write your name and CWID, one digit per box, at the top of every sheet: the sheets are graded separately and matched by CWID.
- Show your work in Parts 2 and 3.

Part 1: Multiple Choice [10 pts]

1 point each. Fill the bubble of the one best answer.

1. Kai runs a food truck. After paying for food, fuel, and the permit, the books show a profit of \$40,000 a year. Kai could instead earn \$55,000 a year as a chef at a restaurant. Kai's economic rent from running the truck is

☐ A ☐ B ☐ C ☐ D

- (A) \$40,000, the profit on the books.
(B) -\$15,000: an economic loss.
(C) \$95,000.
(D) \$15,000.

2. A studio has spent \$2 million on a video game that is half finished, and it cannot get that money back. Finishing the game would cost \$1 million more, and the finished game would bring in \$1.5 million. The studio should

☐ A ☐ B ☐ C ☐ D

- (A) stop, since its total cost of \$3 million is more than \$1.5 million.
(B) stop, since it has already lost \$2 million.
(C) finish it only if it brings in at least \$3 million.
(D) finish it, since \$1.5 million is more than the \$1 million it still costs.

3. In an hour, Ana can bake 6 loaves or 12 muffins, and Raj can bake 2 loaves or 8 muffins. Who has the comparative advantage in muffins?

☐ A ☐ B ☐ C ☐ D

- (A) Ana, since she bakes more muffins.
(B) Raj, since a muffin costs him $\frac{1}{4}$ loaf and costs Ana $\frac{1}{2}$ loaf.
(C) Neither, since Ana has the absolute advantage in both goods.
(D) Ana, since a muffin costs her 2 loaves.

4. A bakery's customers believe they cannot find its sourdough anywhere else. Compared with a bakery whose bread is the same as its rivals', it
- (A) can raise its price and keep many of its buyers.
 - (B) must charge the same price as its rivals.
 - (C) faces no trade-off between price and quantity.
 - (D) loses all its buyers if it raises its price.

☐ A ☐ B ☐ C ☐ D

5. A furniture maker doubles all of its inputs: workers, machines, materials, and space. Its output less than doubles. The firm has
- (A) increasing returns to scale.
 - (B) constant returns to scale.
 - (C) decreasing returns to scale, or diseconomies of scale.
 - (D) economies of scale.

☐ A ☐ B ☐ C ☐ D

6. A print shop pays \$600 a month for its space, and each poster costs it \$2 in paper and ink. What is its average cost per poster if it prints 100 posters a month, and if it prints 300?
- (A) \$8 and \$4.
 - (B) \$2 and \$2.
 - (C) \$6 and \$2.
 - (D) \$8 and \$8.

☐ A ☐ B ☐ C ☐ D

7. A bus company raises its fare from \$2.00 to \$2.20, and daily riders fall from 10,000 to 9,500. The price elasticity of demand is
- (A) 2, so demand is elastic.
 - (B) 0.5, so demand is elastic.
 - (C) 0.5, so demand is inelastic.
 - (D) 5, so demand is elastic.

☐ A ☐ B ☐ C ☐ D

8. A store sells headphones for \$120 a pair to everyone, and each pair costs it \$40. A buyer who values a pair at \$70 does not buy. Selling that buyer one pair at \$60, while everyone else still pays \$120, would
- (A) make the store worse off, since \$60 is below \$120.
 - (B) make the other buyers worse off.
 - (C) add to the deadweight loss.
 - (D) be a Pareto improvement: the buyer gains \$10, the store \$20, and nobody loses.

☐ A ☐ B ☐ C ☐ D

9. Which of these is most likely to be a natural monopoly?
- (A) Restaurants in a large city.
 - (B) The water pipes that serve a town.
 - (C) A drug company that holds the patent on a new medicine.
 - (D) Wheat farms.

☐ A ☐ B ☐ C ☐ D

10. Two gas stations face each other across a road. Their prices are likely to be lowest when
- (A) most drivers compare prices and fill up at the cheaper station.
 - (B) most drivers are loyal to one station and fill up there at any price.
 - (C) drivers see the two stations' gas as different.
 - (D) both stations have low fixed costs.

☐ A ☐ B ☐ C ☐ D

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Sheet 2 of 3. Show your work.

Part 2: Short-Answer Problems [10 pts]

Four questions, 2.5 points each. Answer 1 and 2 in two or three sentences, using the ideas from class; show your work in 3 and 4.

1. [2.5 pts] The owners of a firm bought their factory outright years ago. A new manager argues that since it is paid for, using it costs nothing. What is missing from that argument?

2. [2.5 pts] Cinemas, museums, and software companies routinely charge students less than other customers for exactly the same thing. What is this called, and why do firms do it?

3. [2.5 pts] Economists estimate that, at their profit-maximizing prices, the price elasticity of demand is about 6 for a small economy car and about 3 for a luxury car. Use the markup rule to find each car's markup. What does a markup measure, and why is the luxury car's higher?

4. [2.5 pts] Suppose the price of gasoline rises 25%. Sales of large SUVs fall 10%, and sales of electric cars rise 15%. Find the cross-price elasticity of demand for each with respect to the price of gas, and say what each number tells you about the pair.

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Sheet 3 of 3. Show your work.

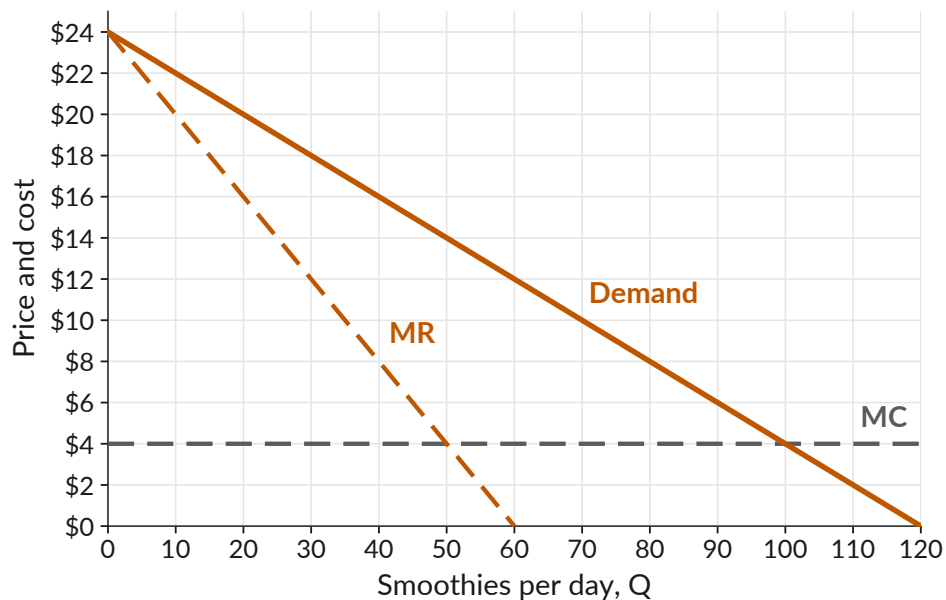
Part 3: Long-Form Question [10 pts]

Lakeside Smoothies

Lakeside Smoothies is a stand at a lake. Its demand curve is

$$Q = 120 - 5P, \quad \text{or equivalently} \quad P = 24 - \frac{Q}{5},$$

where P is the price of a smoothie and Q is smoothies sold per day. Each smoothie costs the stand \$4 in fruit, ice, and labor. The graph plots the demand curve with the stand's marginal revenue (MR) and marginal cost (MC).



(a) [2 pts] On the plot, MR at 50 smoothies is exactly \$4, equal to MC. Check this for the 51st smoothie. Find the price that sells 51 smoothies, calculate revenue at 51 and at 50 smoothies, and take $MR = R(51) - R(50)$. What does your answer tell the stand about selling the 51st smoothie?

(b) [1.5 pts] What are the profit-maximizing quantity and price? Mark that point, E , on the demand curve in the graph above, and draw the price line through it.

(c) [1.5 pts] On the graph, shade and label the consumer surplus, the producer surplus, and the dead-weight loss.

For the rest of the problem, use the areas you shaded on the front of this sheet.

(d) [1.5 pts] Compute the consumer surplus, and explain why that area measures what buyers gain.

(e) [1.5 pts] Compute the producer surplus, and explain why that area measures what the stand gains.

(f) [0.5 pts] Which buyers value a smoothie above its \$4 cost but do not buy at \$14? Give the range, from the ___th to the ___th buyer.

(g) [0.5 pts] Give a price at which selling a smoothie to the 75th buyer would be a Pareto improvement.

(h) [1 pt] Compute the deadweight loss, and explain what it represents.